

1. **What is dynamic binding?**

When a method call is resolved at runtime, it is called dynamic binding.

2. **Define inheritance.**

The process by which one class acquires the properties and methods of another class.

3. **What is the use of switch statement?**

The `switch` statement is used to select one block of code to execute from multiple options.

4. **Define default constructor.**

A constructor without parameters, automatically provided if no constructor is defined.

5. **What is hierarchical inheritance?**

When multiple classes inherit from a single parent class.

6. **List any two access modifiers.**

`public`, `private`

7. **What is interface in Java?**

An interface is a collection of abstract methods that a class can implement.

8. **What is the use of `Math` class?**

The `Math` class provides mathematical functions like `sqrt()`, `pow()`, `abs()`, etc.

9. **Define package.**

A package is a collection of related classes and interfaces grouped together.

10. **What is thread lifecycle?**

The stages of a thread: New → Runnable → Running → Waiting/Blocked → Terminated.

11. **Define message communication.**

Message communication refers to objects interacting by invoking methods of other objects.

12. **What is JVM?**

JVM (Java Virtual Machine) is a software engine that runs Java bytecode and makes Java programs platform-independent.

13. **Define polymorphism.**

Polymorphism means the ability of one task to be performed in different ways (e.g., method overloading and overriding).

14. **What are tokens in Java?**

Tokens are the smallest elements in Java: keywords, identifiers, literals, operators, and separators.

15. **What is type casting?**

Converting one data type into another is called type casting (e.g., `int x = (int) 10.5;`).

16. **Define abstract class.**

A class declared with the `abstract` keyword that may have abstract methods is called an abstract class.

17. **What is method overloading?**

Defining multiple methods with the same name but different parameter lists.

18. **List any two classes of `java.lang` package.**

`String`, `Math`

19. **Define multiple inheritance in Java.**

When a class can inherit behavior from more than one source. In Java, achieved using **interfaces**.

20. **What is final keyword in Java?**

`final` is used to declare constants, prevent method overriding, or inheritance of a class.

21. **Define encapsulation.**

Wrapping data and methods into a single unit (class) and controlling access using access modifiers.

22. What is multithreading?

Multithreading allows a program to perform multiple tasks at the same time using threads.

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